

Rajentharr Jeganathan

SOFTWARE ENGINEER

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SUMMARY

Accomplished Software Engineer with over six years of experience building robust enterprise applications across aviation systems, real-time risk management, and e-commerce platforms. Proficient in Agile, Sprint, and Scrum methodologies, with a strong emphasis on process excellence and team collaboration. Skilled in full-stack development, database design, event-driven and distributed system design — sagas, CQRS, change data capture, and idempotent transaction processing — and technical documentation, with a proven track record of delivering high-quality solutions. Recognised for being self-motivated, trustworthy, and highly adaptable in dynamic and collaborative work environments.

CAREER HIGHLIGHTS

- Developed and maintained web applications using Java, Spring, and MySQL.
- Built internal APIs and batch applications leveraging Spring Boot, Scala, Hive, MongoDB, and MySQL.
- Developed Spring Batch jobs for large-scale data processing and transformation.
- Deployed and scheduled batch jobs using Azkaban for efficient pipeline management.
- Created automation scripts and support tools using Shell and Python to streamline development and operations.
- Collaborated in an Agile development environment, using Jira for sprint management and Bitbucket for version control.
- Enhanced code quality by integrating SonarQube and performed performance testing with Taurus.
- Designed and managed database schemas, ensuring data integrity and operational efficiency.
- Authored and reviewed technical documentation, including Proof of Concept (POC) reports and release notes.
- Handled live issue resolution, ensuring consistent application stability and uptime.
- Used tools including Postman, MySQL Workbench, Docker, and Localstack (for local AWS environments) in development and testing workflows.

TECHNICAL SKILLS

Programming · Java, Python, PHP, JavaScript, Scala

Databases · MySQL, MongoDB, Postgres, Hive

Web · React, HTML, CSS, Ag-Grid, Apex-charts, Material UI

Frameworks · Spring Boot, Spring Batch, Spring, Android, Apache Kafka, Redpanda, Redis

Distributed Systems · Event-Driven Architecture, Saga Orchestration, CQRS, Change Data Capture, Transactional Outbox, Idempotency

Testing · JUnit, Mockito, Testcontainers

Version Control · Git (Bitbucket / GitHub)

Cloud · AWS

Tools · Jira, MySQL Workbench, Postman, IntelliJ IDEA, SonarQube, Redis Desktop Manager, Docker, Kubernetes, Jenkins, Kibana, Taurus, Debezium, OpenTelemetry, Grafana, Hadoop toolsets (Sqoop, Beeline, Hive)

CORE COMPETENCIES

- Software and Application Design
- Technical Architecture Design and Documentation
- Agile Methodology
- Research and Development
- Distributed Systems Design
- Event-Driven Architecture

EXPERIENCE

Software Engineer — Rakuten

Jan 2023 – Present

Rakuten is an internet company specialising in high-performance e-commerce applications, mobile networks, banking, and other financial services.

- Performed performance testing using Taurus and Gatling.
- Coordinated end-to-end design, from schema to technical implementation.
- Provided production support for releases, alerts, and schema changes.
- Optimized application performance and implemented graceful shutdown handling.
- Utilized AI tooling to streamline work across the software development lifecycle, with all AI-assisted output independently reviewed and confirmed.
- Authored documentation including release and rollback manuals, API specifications, design documents, and Architecture Decision Records (ADRs).
- Mentored junior engineers and new team members.

Software Engineer — ZeroBeta

Jan 2021 – Dec 2022

Specialised in delivering high-performance computing products and solutions tailored for financial risk management.

- Developed and maintained modern applications using Java, Spring Boot, and MySQL.
- Designed and implemented frontend features with React, Ag-Grid, Apex-charts, Material-UI, and Redux.
- Integrated caching solutions with Redis and implemented message passing via Kafka.
- Participated in Agile sprint planning and version control using Jira and Bitbucket.
- Improved and maintained code quality through SonarQube integration and continuous improvement practices.
- Conducted requirements gathering, POC development, technical feasibility analysis, and performance impact assessments.
- Authored and maintained technical documentation, including database design and data manipulation procedures.
- Provided live issue resolution to support application stability and production environment reliability.

Software Engineer — Information Systems Associates (Accelaero)

Feb 2020 – Jan 2021

Leading provider of advanced technology solutions for the global travel and aviation industry since 2005.

- Developed modern applications using Java, Spring Boot, Protobuf, ANTLR, and PostgreSQL.
- Contributed to frontend development with React and Redux.
- Implemented APIs and business logic, ensuring high JUnit test coverage and maintaining code quality through SonarQube.
- Utilized Jira for sprint management and Bitbucket for version control in an Agile environment.

Software Engineer Intern — GE Transportation Company

Jun 2018 – Dec 2018

Global leader in equipment, systems, digital solutions, and services for freight rail, transit, mining, industrial, and marine sectors.

- Developed and debugged features for products built with Spring and Hibernate.
- Created automation test scripts using Serenity/JS to ensure functional reliability.
- Explored machine learning concepts and implemented a sample K-means clustering algorithm.
- Participated in Agile practices and actively contributed to code reviews to improve product quality.

PERSONAL PROJECTS

Lynx — Multi-Currency Ledger Engine · Personal project

May 2026 – Present

A distributed ledger system for cross-currency transfers, built to production patterns: saga orchestration, event-driven CQRS, and end-to-end idempotency.

- Designed an append-only, double-entry ledger in PostgreSQL at SERIALIZABLE isolation as the system's single source of truth, with every write balanced and nothing ever updated or deleted.
- Implemented a 3-step saga orchestrator (hold → FX rate lock → settle) with compensating actions on failure, and a recovery worker that resumes orphaned sagas after a crash using FOR UPDATE SKIP LOCKED.

- Guaranteed exactly-once transfers by layering an idempotency key, a deterministic saga identifier, and a database UNIQUE constraint, so retries and crashes cannot produce duplicate ledger entries.
- Published domain events through a transactional outbox streamed off the Postgres write-ahead log by Debezium CDC, ensuring events can never be lost or orphaned from the writes that produced them.
- Built CQRS read projections for balances and transaction history so read traffic never touches the ledger write path.
- Secured service-to-service calls with service-identity tokens carrying on-behalf-of user context, per-client circuit breakers, and correlation IDs traced end-to-end through OpenTelemetry.
- Covered the system with automated tests using JUnit and Testcontainers against a real PostgreSQL instance, and documented each trade-off in Architecture Decision Records.

Java 21 · Spring Boot 3.3 · PostgreSQL · Redis · Kafka/Redpanda · Debezium · OpenTelemetry · Docker · Testcontainers

EDUCATION

BSc. Eng (Hons), Computer Science and Engineering

Nov 2015 – Jan 2020

Faculty of Engineering, University of Moratuwa, Colombo, Sri Lanka